

LONG VOWEL IN ESTONIAN OLD TEXT: A CORPUS STUDY

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Abstract. The development of Estonian orthography has taken an unconventional trajectory. Early Estonian writers used a Finnish-based system, which was later modified by Lutheran missionaries. German and Swedish letters also left a mark on orthography for decades, until the script was standardized in the 19th century. Regarding long vowels, linguists like Hermann introduced innovations such as distinctions between short and long vowels, along with the grave accent. But there are other ways adopted too. Studying these old texts and getting insights into how Estonian orthography developed is interesting, mainly because of the variations that can be seen. Additionally, using a corpus provides a data-driven perspective to the analysis.

1. Introduction

This paper aims to study the orthographic representation of vowel length in old written Estonian texts. The research questions and methods of analysis are presented in Section (3). Section (4) discusses the operational definition. Section (5) presents findings and conclusions. To begin with, section (2) will describe known facts about Estonian long vowels in written form.

2. Long vowels in old Estonian texts

Long vowels are characterized by their extended pronunciation. Modern Estonian uses double letters to show long vowels (Erelt, 2007, p. 8). However, old Estonian texts used an ad hoc orthography that was influenced by Latin, Finnish, and German writing systems. Long vowels in open syllables were often represented using a single letter, e.g., *loma* (*looma* 'to create'), *ma* (*maa* 'earth'), *se* (*see* 'this'), *ramat* (*raamat* 'book'). In a closed syllable, two letters are used for a long vowel, e.g., *malt* 'from the country,' *siis* 'then,' *juuste* 'hair.' Additionally, grave accents (à, è) were also used (Erelt, 2007, p. 288).

3. Research question and method

The study aimed to answer the question, of how long vowels are represented in old Estonian texts corpus. To achieve this objective, this study uses text from the Estonian old Written Language Corpus (<http://vakk.ut.ee>). It is a collection of texts that date back to the 15th to 19th centuries. The corpus contains 3,016 texts and a total of 3,322,111 words, of which 2,846,263 are Estonian words. Researchers manually marked the texts from the 15th, 16th, and 17th centuries. The texts from the 18th century were marked with an auto tagger (Kulli, 2023). The corpus is updated regularly. The most recent version (as of November 6,

2023) is used in this study. The algorithm for data refinement is developed in Python using regex which creates analytical data (for more information refer to: *vowel_length.py*). The data analysis is done manually on analytical data. The investigation process involves three stages: data collection, data cleaning, and data analysis. For vowel comparison, both modern spelling (*märksõna*) and older spelling (*sõne*) are considered. The assumption is that the modern spelling in the corpus is accurate. The verification of modern spelling falls beyond the scope of this study.

3.1 Data collection

The first step in the analysis involved collecting data. The corpus filtered to include texts written in Estonian in the islands, Tallinn, and Tartu regions. This resulted in a dataset of 1,875,119 rows.

3.2 Data cleaning

The study uses a new spelling (*märksõna*) as the basis for analysis. The assumption is new spelling is in the correct form. The following criteria are used to streamline the dataset:

- 3.2.1 Data with NULL in '*märksõna*' was ignored.
- 3.2.2 Only data with long vowels is considered.
- 3.2.3 Some of the new and old spelling data contained special characters, ("_", "+", "-", "=", "/", "\"", "[", "]", ":", "{x}", "{l}", "") making it not suitable for analysis. Such characters are removed.
- 3.2.4 Since one lemma had multiple rows, the combination of '*märksõna*' and '*sõne*' was compared to eliminate duplicates.

Moreover, an important question that needs to be tackled is how to compare the two orthographies effectively. A general observation is that long vowels are often represented by a double letter in the new Estonian orthography. On the other hand, old orthography is ad hoc. This makes direct vowel-to-vowel comparison impractical. To overcome this challenge, a solution was used to find the position of the long vowel in the new orthography. This resulted in a new column named '*long_vowel_position*.' This information was used to extract the corresponding long vowel from old orthography into column '*sõne_long_vowel_extract*' and the exact long vowel from newer spelling into column '*märksõna_long_vowel*.' This process yielded a dataset of 8,457 rows.

Given the large size of the dataset, manual processing would be impractical. To address this limitation, the option is considered for further data reduction. The solution is only distinct rows based on

'marksona_long_vowel' and *'sona_long_vowel_extract'* are kept. This method resulted in a reduced dataset of 459 rows.

3.3 Data analysis

The final analytical dataset, comprising 459 rows, was analyzed manually. The analysis involved comparing the long vowel representation in the old and new orthographies using the operational definition in section 4. This allowed for the identification of the most common patterns of long vowel representation in old Estonian texts.

4. Operational definition

The operational definition serves as a set of guidelines for data analysis, and it is applied to an analytical dataset of 459 records. Here are the terms used in the operational definition.

	Description	Example
S1	New spelling	hooreb
S2	Old spelling	horeb
VV1	Long vowel from S1	oo
VV2	Long vowel from S2	or
V1	Vowel from V1	o
V2	Vowel V2	o

Table 1: Abbreviations used in the operational definition.

4.1 [VV1 == VV2]

(Here: VV1 is *'marksona_long_vowel'* and VV2 is *'sona_long_vowel_extract'*.)

e.g.: 'üü' == 'üü'

Action: Note VV1.

4.2 [VV2.contains(V1)]

(Here: vowel V1 is from column *'marksona_long_vowel'* and VV2 is *'sona_long_vowel_extract'*)

e.g.:

- 'üg' contains 'ü.'

- 'gü' contains 'ü.'

- 'üi' contains 'ü.'

Action: Note V1.

4.3 [not VV2.contains(V1)]

(Here: Vowel V1 is from column 'marksona_long_vowel' and VV2 is 'sona_long_vowel_extract')

e.g.:

- 'bg' does not contain 'ü.'
- 'ag' does not contain 'ü.'
- 'gi' does not contain 'ü.'

Action:

- Note corresponding S1 and S2
- Add NV (Not a vowel) flag, for further analysis.

(Here S1 is column 'märksõna' and S2 is column 'sõne')

5. Results and discussion

Following the application of the operational definition, the results are collected.

Description	Rows
Total	459
Results Collected	370
Problematic Cases	89

Table 2: Result of data analysis

These 370 rows represent instances where the long vowel representations align with the specified parameters.

		Variations														
Long Vowels	aa	a	aa	e	oa	ää	åå	ho	ä	a'	aâ	o	ö	ä		
	ää	ää	eh	ee	äh	eä	æ	ä	e	äe	æe	ê	a	èè	ea	æ
	ee	ä	æ	äe	e	ê	ẽ	ee	eê	eê	èè	êe	ẽe	êe	i	ie
	ii	e	i	j	y	ÿ	yy									
	oo	e	o	ô	ö	oo	oò	oô	u							
	öö	o	ö	öö	ü											
	õõ	ee	o	ö	oö	öö										
	uu	oô	õt	u	ü	ue	uo	uu	üu	v	vw	w				
	üü	i	u	ü	üü	y	ÿ	ÿÿ								

Table 3: Variations in long vowel representation in old Estonian text.

The results showed that there were a variety of ways to represent long vowels, including using double letters, single letters, grave accents, diaeresis, macron, circumflex, etc. The use of these diacritics was not always consistent in old Estonian texts. There are many cases where the same long vowel is represented in different ways, depending on the author or the region where the text was written.

Further, problematic cases are identified, and these instances are flagged as 'NV' to denote potential irregularities or complexities in the representation of long vowels. A total of 89 rows falls into this category. This data requires further investigation or expert analysis. Here are a few examples from this category to prompt further investigation:

Category	Example
No Vowel	missing a vowel sound. e.g., ' <i>Jernsalemmil</i> ' corresponding standard spelling ' <i>Jeruusalemm.</i> '
Alternate Spellings	e.g., ' <i>kz, ks = kaas</i> '
Semantically Ambiguous Representations	Some cases appear semantically ambiguous. e.g., ' <i>eph = jääma</i> ,' ' <i>naist = need</i> ,' ' <i>Wihendes = viies</i> ,' ' <i>wyhme = viima</i> ,' ' <i>Ioket = jooma</i> ,' ' <i>sewde = sööma</i> '.
Potential Abbreviations	e.g., ' <i>Al = halleluuja</i> ,' whether potential abbreviations or shorthand forms of ' <i>halleluuja</i> '?

Table 4: Identified problematic cases with examples.

6. Conclusion

This paper has investigated the representation of long vowels in old Estonian texts using the Estonian old written language corpus. The findings revealed a variety of ways used for representing long vowels. There are a few problematic cases where it is difficult to find the long vowel in old texts. These cases necessitate further investigation to establish the appropriate long vowel representation.

Terms

VAKK	Vana Kirjakeele Korpus
Märksõna	Modern spelling
Sõne	A text word exactly as it is used in the old text

Supplement files

1. Python script: vowel_length.py
2. Data file used for analysis: long-vowels-wip-459-rows.xlsx.

References

Erelt, M. (2007). *Estonian language* (Second Edition, Vol. 1). Estonian Academy Publishers.

Kulli, P. (2023). *Vana kirjakeele korpus*. <https://vakk.ut.ee>